EPDM rubber is inherently poor in oil resistance because its non-polar, saturated hydrocarbon backbone has no affinity for hydrocarbon oils—it swells readily when exposed to paraffinic or aromatic hydrocarbons . Several strategies can overcome this limitation, each with distinct mechanisms and trade-offs.

□ Blending with Polar Polymers

The most straightforward approach is blending EPDM with oil-resistant polar elastomers.

Polysulfide rubber and chlorosulfonated polyethylene (CSM) blends: A patented high oil-resistant EPDM gasket formulation combines EPDM (51–63 parts) with polysulfide rubber (38–52 parts) and CSM (22–34 parts) . Polysulfide rubber introduces polar sulfide linkages that repel non-polar hydrocarbons, while CSM contributes chlorine-containing polar groups that reduce oil absorption. This multi-polymer blend reportedly achieves excellent oil resistance while maintaining good air tightness and temperature resistance .

Advantages: Relatively simple compounding; leverages commercial polymers.

Limitations: Compatibility between phases must be managed; mechanical properties may be compromised without proper compatibilization.

□ Radiation Curing with Crosslinking Monomers

A more sophisticated approach involves radiation curing EPDM in the presence of acrylic crosslinking monomers rather than conventional sulfur vulcanization.

Mechanism: When EPDM is blended with crosslinking monomers such as trimethylolpropane trimethacrylate (TMPT) or triallyl isocyanurate, radiation curing creates unusual crosslinking species with multiple points of unsaturation . The acrylic monomer first forms oligomers with reactive unsaturation sites, which then crosslink the EPDM matrix. This creates a denser, more oil-repellent network than sulfur curing can achieve .

Performance data: Radiation-cured EPDM formulations achieve volume swell of approximately **90–110%** in ASTM #3 oil after 70 hours at 150°C, compared to traditional sulfur-cured EPDM which exhibits very poor resistance . Critically, this improvement is specific to EPDM—radiation curing does not improve oil resistance of NBR or polyacrylate .

Formulation components (per 100 parts EPDM): 18–22 parts crosslinking monomer, 7–9 parts chlorinated or brominated organic additive (e.g., Chlorowax LV), 9–11 parts hindered phenolic antioxidant (e.g., Irganox 1010), 10–15 parts zinc oxide, and 5–8 parts magnesium oxide, cured with dicumyl peroxide .

Advantages: Achieves both exceptional heat resistance and improved oil resistance in a single EPDM formulation .

Limitations: Requires electron beam radiation equipment; limited to specific crosslinking monomer systems.

□ Grafting Polar Chains onto EPDM Backbone

Radiation-induced grafting of acrylonitrile (ACN) onto EPDM introduces polar nitrile groups directly onto the rubber backbone .

Mechanism: Acrylonitrile monomers are grafted onto EPDM chains via mutual radiation grafting. The nitrile groups (−C≡N) are highly polar and incompatible with non-polar hydrocarbon oils, dramatically reducing oil swelling. The grafted copolymer also exhibits better compatibility with polar polymer matrices .

Optimization data: The optimum solvent composition for maximum grafting efficiency is **acetone:CCl_4 = 20:80**. Degree of grafting increases with radiation dose and monomer content, but decreases with dose rate .

Advantages: Directly modifies EPDM molecular structure; no phase separation issues.

Limitations: Requires radiation source; grafting efficiency depends on precise control of solvent, dose, and monomer concentration.

□ Blending with HNBR/PVC Systems (For Reference)

While not an EPDM modification per se, understanding NBR-based oil-resistant systems provides useful formulation guidance. Partially hydrogenated NBR with 45–50% acrylonitrile content blended with PVC (50/50 ratio) achieves an excellent balance of oil resistance, cold resistance, and dynamic properties . Higher acrylonitrile content improves oil resistance but sacrifices low-temperature flexibility . PVC addition enhances hardness, tensile strength, and solvent resistance in NBR systems .

□ Optimum Combination Ratios

Based on available data, the following formulation ratios are recommended for oil-resistant EPDM systems:

Strategy	Key Components	Optimum Ratio (parts by weight)	Oil Swell Target
Polymer Blend	EPDM / Polysulfide / CSM	51–63 / 38–52 / 22–34	Not specified
Radiation Cure	EPDM / TMPT / Chlorowax / Antioxidant	100 / 18–22 / 7–9 / 9–11	~90–110% (ASTM #3, 70h@150°C)
ACN Grafting	EPDM / ACN monomer	Variable (dose-dependent)	Reduced swelling (qualitative)

□ Practical Selection Guidance

For **sealing applications requiring the best oil resistance**, radiation-cured EPDM with TMPT crosslinking monomer offers the most proven performance, achieving oil swell below 130% while maintaining heat resistance to 177°C . For **cost-sensitive applications**, the EPDM/polysulfide/CSM blend provides a simpler compounding route . For **applications where EPDM must be used but oil resistance is secondary**, ACN grafting offers a targeted molecular modification .

It is important to note that EPDM, even when modified, will not match the inherent oil resistance of NBR, HNBR, or FKM. HNBR with high acrylonitrile content offers superior oil resistance with a TR10 of −37°C , while FKM provides the broadest chemical resistance but with poor low-temperature flexibility . The selection ultimately depends on the specific balance of oil resistance, temperature range, and cost required for the application.